

SUPERELLIPTIC SELMER GROUPS

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1. THE PROBLEMS

Definition 1.0.1. Let k be a field. (♠ TODO: projective closure, plane curve, affine equation) (♠ TODO: smooth completion of an affine curve) A *superelliptic curve over k* is the smooth completion of the plane curve with affine equation of the form

$$y^d = f(x)$$

where $f(x) \in k[x]$ and $d \geq 2$ is some integer.

Definition 1.0.2. Let k be a field. Let C be a superelliptic curve (Definition 1.0.1) over k given as the smooth completion of the plane curve with affine equation

$$y^d = f(x)$$

where $f(x) \in k[x]$ and $d \geq 2$ is some integer. Given an element $u \in k^\times$, the *μ_d -twist of C* is the superelliptic curve given as the projective closure of the plane curve with affine equation

$$y^d = \frac{1}{u} f(x).$$

Here is the main problem:

Context 1.0.3. Let $K = \mathbb{F}_q(t)$ be a global function field of characteristic coprime to 6. Let $\ell \geq 5$ be a prime number such that $\gcd(\ell, q) = 1$. Write $F_{n,\ell}$ for the set of ℓ th power free polynomials over $\mathbb{F}_q$ of degree n . Let v be a prime number coprime to $6q\ell$.

Context 1.0.4. Assume Context 1.0.3. Fix a superelliptic curve (Definition 1.0.1) C/K whose affine equation is given by

$$C : y^\ell = f(x)$$

for some non-trivial polynomial $f(x) \in K[x]$.

Given an ℓ -th power free polynomial $g \in \mathbb{F}_q[t]$ write C_g for the μ_ℓ -twist (Definition 1.0.2) of C by g . In other words C_g is the smooth completion of the plane curve with affine equation of the form

$$C_g : gy^\ell = f(x)$$

Problem 1.0.5. Assume Context 1.0.3 and Context 1.0.4

For each integer $j \geq 0$, what is the asymptotic value of the quantity

$$\frac{\#\{g \in F_{n,\ell} : \dim_{\mathbb{F}_v} \text{Sel}_v(\text{Jac}(C_g)/K) = j\}}{\#F_{n,\ell}}$$

(Definition D.0.2) (Definition A.1.3) as $n \rightarrow \infty$?

Here are subproblems

Problem 1.0.6.

Problem 1.0.7. Assume Context 1.0.3.

Write $\mathcal{J}_g$ for the Néron model of $\text{Jac}(C_g)$ over

APPENDIX A. ABELIAN VARIETIES

Definition A.0.1. Let k be a field. A *(algebraic) variety over k* is an integral, separated scheme of finite type over k .

Definition A.0.2. Let S be a scheme. An *algebraic group scheme over S* (or an *S -group scheme*) is a group object G in the category of schemes over S ; that is, G is an S -scheme equipped with S -morphisms: $m : G \times_S G \rightarrow G$ (*multiplication*), $i : G \rightarrow G$ (*inverse*), and $e : S \rightarrow G$ (*identity/identity section*), satisfying the group axioms expressed by the commutativity of the following diagrams:

$$\begin{array}{lcl}
 \text{1. Associativity} & \begin{array}{ccc} G \times_S G \times_S G & \xrightarrow{m \times \text{id}} & G \times_S G \\ \text{id} \times m \downarrow & & \downarrow m \\ G \times_S G & \xrightarrow{m} & G \end{array} & \\
 \text{2. Identity} & \begin{array}{ccc} G \times_S S & \xrightarrow{\text{id} \times e} & G \times_S G \\ \searrow \cong & \downarrow m & \\ & G & \end{array} & \begin{array}{ccc} S \times_S G & \xrightarrow{e \times \text{id}} & G \times_S G \\ \searrow \cong & \downarrow m & \\ & G & \end{array}
 \end{array}$$

3. **Inverse**
$$\begin{array}{ccc} G & \xrightarrow{(\text{id}, i)} & G \times_S G \\ \text{id} \downarrow & & \downarrow m \\ G & \xrightarrow{e \circ \pi} & G \end{array}$$
 where $\pi : G \rightarrow S$ is the structure morphism and $e \circ \pi$ sends g to the identity section.

If G is affine over S , we call it an *affine group scheme over S* .

If the base scheme S is the spectrum of a field k , then we call G a *k -algebraic group* or an *algebraic group (scheme) over k* . If G is additionally a k -variety, then we call G a *k -group variety*.

Definition A.0.3. Let S be a scheme, and let G and H be S -algebraic groups (Definition A.0.2). A morphism of S -schemes $f : G \rightarrow H$ is a *homomorphism of algebraic groups* if f is a group homomorphism, i.e.,

- $f(m_G(x, y)) = m_H(f(x), f(y))$ for all $x, y \in G$,
- $f(i_G(x)) = i_H(f(x))$ for all $x \in G$, and
- $f(e_G) = e_H$.

It is called an *isomorphism of algebraic groups (over S)* if it is additionally an isomorphism of S -schemes. If there exists an isomorphism $f : G \rightarrow H$ of algebraic groups over S , then G and H are said to be *isomorphic S -algebraic groups*.

Definition A.0.4 (Kernel of a homomorphism of group schemes over a scheme). Let S be a scheme and let $\phi : G_1 \rightarrow G_2$ be a homomorphism of algebraic group schemes over S (Definition A.0.3).

The *kernel of ϕ* , denoted by $\ker \phi$ or $A[\phi]$, is the fiber of ϕ at the identity section (Definition A.0.2) $e_2 : S \rightarrow G_2$ of G_2 . Equivalently, $\ker \phi$ is the subgroup scheme of G_1 defined as the pullback:

$$\ker \phi = G_1 \times_{G_2} S,$$

(♠ TODO: finite group scheme) where the map $G_1 \rightarrow G_2$ is ϕ and the map $S \rightarrow G_2$ is the identity section. If ϕ is a finite morphism, then $\ker \phi$ is a finite group scheme over S .

(♠ TODO: group variety)

Definition A.0.5 (Abelian variety over a field). Let k be a field. An *abelian variety over k* is a complete, connected algebraic group variety defined over k , i.e.

- A is a smooth, proper, geometrically connected algebraic variety over k ,
- A is endowed with a group structure defined by morphisms of varieties over k (multiplication $m : A \times_k A \rightarrow A$ and inverse $i : A \rightarrow A$),
- the group law satisfies the group axioms scheme-theoretically.

In particular, an abelian variety is a projective algebraic group variety (Definition A.0.2) over k .

Definition A.0.6 (Isogeny of abelian varieties over a field). (♠ TODO: finite group scheme) Let k be a field and let A and B be abelian varieties (Definition A.0.5) over k . An *isogeny of abelian varieties over k* is a homomorphism (Definition A.0.3) $\varphi: A \rightarrow B$ of group schemes over k that is surjective and has finite kernel (Definition A.0.4). The kernel $\ker \varphi$ is a finite group scheme over k (Definition A.0.4). Equivalently, φ is a finite, faithfully flat homomorphism of abelian varieties over k .

A.1. Picard groups and Jacobians.

Definition A.1.1. (♠ TODO: picard group of an algebraic space) Let X be a scheme. The *Picard group of X* , denoted by $\text{Pic}(X)$, is the group of isomorphism classes of invertible sheaves (line bundles) on X (with respect to the Zariski topology), with the group operation induced by the tensor product of sheaves.

More precisely:

1. Two invertible sheaves $\mathcal{L}$ and $\mathcal{M}$ are isomorphic if there exists an $\mathcal{O}_X$ -module isomorphism $\phi: \mathcal{L} \xrightarrow{\sim} \mathcal{M}$.
2. The tensor product of invertible sheaves induces a well-defined group operation on the set of isomorphism classes. (♠ TODO: sheafy dual)
3. The inverse of an invertible sheaf $\mathcal{L}$ is given by its dual $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$.
4. The identity element is the class of the structure sheaf $\mathcal{O}_X$.

(♠ TODO: class group of ring) When $X = \text{Spec}(A)$ is an affine scheme, there is a canonical isomorphism

$$\text{Pic}(\text{Spec}(A)) \cong \text{Cl}(A),$$

where $\text{Cl}(A)$ is the class group of the ring A (the group of isomorphism classes of rank-1 projective modules over A , also known as the Picard group of the ring).

(♠ TODO: extract this out) If X is a Noetherian, integral, and locally factorial scheme, then $\text{Pic}(X)$ is isomorphic to the Weil divisor class group $\text{Cl}(X)$.

Definition A.1.2. Let X/k be a smooth, geometrically integral projective variety over a field k .

The relative Picard functor $\text{Pic}_{X/k}$ is representable by a smooth scheme over k . (♠ TODO: show that the identity component is an abelian variety) The identity component $\text{Pic}_{X/k}^0$ is in fact an abelian variety and is called the *Picard variety of X/k* .

Definition A.1.3 (Jacobian variety of a curve). (♠ TODO: separate out jacobian of just a proper curve and the jacobian of a nice curve) Let X be a proper curve over a field k .

1. The relative Picard functor $\text{Pic}_{X/k}$ is representable by a smooth group scheme (Definition A.0.2) over k . The *(generalized) Jacobian of X* refers to the identity component $\text{Pic}_{X/k}^0$ of this group scheme.
2. Now assume that X is smooth, is projective, and has geometrically connected components. The Jacobian is an abelian variety (Definition A.0.5) over k and

may be called the *Jacobian variety of X* . It may be denoted by notations such as $\text{Jac}(X)$, Jac_X , or J_X .

Under these assumptions, note that $\text{Jac}(X)$ coincides with the Picard variety (Definition A.1.2) of X/k . Equivalently, J_X parametrizes isomorphism classes of degree-zero line bundles on X , where a line bundle $\mathcal{L}$ has *degree zero* if its degree (computed as the sum of orders of any rational function defining it) vanishes.

The Jacobian variety $\text{Jac}(X)$ has dimension g where g is the geometric genus of X .

When X is equipped with a base point, the Jacobian variety carries a canonical principal polarization Θ , making (J_X, Θ) a principally polarized abelian variety.

APPENDIX B. NÉRON MODELS OF ABELIAN VARIETIES

Definition B.0.1 (Dedekind domain). An integral domain R is called a *Dedekind domain* or a *Dedekind ring* if it satisfies the following equivalent conditions:

- R is Noetherian, integrally closed in its field of fractions, and every nonzero prime ideal of R is maximal.
- Equivalently: for every nonzero prime ideal $\mathfrak{p}$ of R , the localization $R_{\mathfrak{p}}$ is a discrete valuation ring.

Definition B.0.2 (Dedekind scheme). A *Dedekind scheme* is a scheme S that is integral, Noetherian, normal, and of Krull dimension 1. Equivalently, every point $s \in S$ has an open neighborhood $U = \text{Spec } R$ such that R is a Dedekind domain (Definition B.0.1). In particular, for every point $s \in S$, the local ring $\mathcal{O}_{S,s}$ is either a field (if s is the generic point) or a discrete valuation ring (if s is a closed point).

Definition B.0.3 (Special fiber and generic fiber). Let S be a integral scheme, let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let $s \in S$ be a closed point.

1. For any morphism of schemes $f : X \rightarrow S$, the *generic fiber of X over S* (or simply the *generic fiber*) is the base change:

$$X_{\eta} = X \times_S \text{Spec } K$$

of f along $\text{Spec } K \rightarrow S$.

2. For any morphism of schemes $f : X \rightarrow S$, the *special fiber of X at s* (or simply the *special fiber*) is the base change:

$$X_s = X \times_S \text{Spec } k(s)$$

of f along $\text{Spec } k(s) \rightarrow S$, where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field at s .

Definition B.0.4 (Néron model of an abelian variety). Let S be a Dedekind scheme (Definition B.0.2), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let A_K be an abelian variety over K (Definition A.0.5).

A *Néron model of A_K over S* is a smooth separated group scheme $\mathcal{A}$ of finite type over S whose generic fiber (Definition B.0.3) $\mathcal{A}_{\eta}$ is isomorphic to A_K , satisfying the following

universal property (the **Néron mapping property**): for every smooth S -scheme U and every K -morphism $f_K : U_\eta \rightarrow A_K$, there exists a unique S -morphism $f : U \rightarrow \mathcal{A}$ extending f_K .

If such a model exists, it is unique up to unique isomorphism.

APPENDIX C. GALOIS COHOMOLOGY

Definition C.0.1. Let G be a group and R a not necessarily commutative ring. Let A be a left/right $R[G]$ -module.

1. The **cohomology group of G with coefficients in A** or the **group cohomology of A as an $R[G]$ -module**, denoted $H^n(G, A)$ or $H^n(G; A)$ for $n \geq 0$, is the n -th right derived functor object $(R^n(-^G))(A)$ of the invariants functor $A \mapsto A^G$, which is left exact by ??.
2. The **homology group of G with coefficients in A** or the **group homology of A as an $R[G]$ -module**, denoted $H_n(G, A)$ or $H_n(G; A)$, for $n \geq 0$, is the n -th left derived functor object $(L_n(-_G))(A)$ of the coinvariants functor $A \mapsto A_G$, which is right exact by ??.

Definition C.0.2. Let R be a topological ring and G a topological group. Let $R\text{-}\mathbf{TopMod}_G$ be the category of (left) topological G -modules over R .

Let M be an object in $R\text{-}\mathbf{TopMod}_G$. The **continuous cochain complex** $C_{\text{cont}}^\bullet(G, M)$ is defined as follows:

1. $C_{\text{cont}}^n(G, M) = \text{Map}_{\text{cont}}(G^n, M)$, the R -module of continuous maps from G^n to M .
2. The **coboundary map** $d^n : C_{\text{cont}}^n(G, M) \rightarrow C_{\text{cont}}^{n+1}(G, M)$ is given by:

$$(d^n f)(g_1, \dots, g_{n+1}) = g_1 f(g_2, \dots, g_{n+1}) + \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) + (-1)^{n+1} f(g_1, \dots, g_n)$$

Definition C.0.3. Let R be a topological ring and G a topological group. Let $R\text{-}\mathbf{TopMod}_G$ be the category of (left) topological G -modules over R .

The **continuous cohomology groups** $H_{\text{cont}}^n(G, M)$ are defined as the cohomology groups of the continuous cochain complex (Definition C.0.2) $C_{\text{cont}}^\bullet(G, M)$.

Theorem C.0.4. (♠ TODO: verify these facts)

Let R be a topological ring and G a topological group. Let $R\text{-}\mathbf{TopMod}_G$ be the category of (left) topological G -modules over R .

Consider the G -invariant functor $(-)^G : R\text{-}\mathbf{TopMod}_G \rightarrow R\text{-}\mathbf{TopMod}$ (??).

Assume either of the following:

1. G is profinite, R is discrete, and M is discrete; in this case, by the “right derived functors of $(-)^G$ ”, we mean the right derived functor on the category of discrete G -modules, which has enough injectives (♠ TODO: cite; apparently the category of discrete G -modules over a topological ring R where G is profinite is an abelian category with enough injectives).

2. G is locally compact, and M is complete; in this case, by the “right derived functors of $(-)^G$ ”, we mean the right derived functor in the sense of relative homological algebra, using resolutions that split as sequences of topological spaces (♠ TODO: figure out what this means).

The continuous cohomology groups (Definition C.0.3) $H_{\text{cont}}^n(G, M)$, i.e. the cohomology of the continuous cochain complex (Definition C.0.2) $C_{\text{cont}}^\bullet(G, M)$, is naturally isomorphic to the the right derived functors of $(-)^G$. In other words,

$$H^n(C_{\text{cont}}^\bullet(G, M)) \cong (R^n(-^G))(M)$$

Definition C.0.5. Let R be a topological ring (e.g., a ℓ -adic ring $\mathbb{Z}_\ell$) (♠ TODO: ell-adic ring).

1. Let L/K be a Galois field extension. Let M be a topological G -module over R . The *Galois cohomology group of L/K with coefficients in M* is:

$$H^n(L/K, M) = H_{\text{cont}}^n(\text{Gal}(L/K), M)$$

(Definition C.0.3).

2. Let K be a field with absolute Galois group $G_K = \text{Gal}(\bar{K}/K)$. Let M be a topological G -module over R .

The *Galois cohomology group of K with coefficients in M* is:

$$H^n(K, M) = H^n(\bar{K}/K, M) = H_{\text{cont}}^n(\text{Gal}(\bar{K}/K), M).$$

In either case, one also lets $H^n(G, M)$ denote the Galois cohomology groups where G is the relevant Galois group.

This framework is used to define ℓ -adic cohomology, where M is often a $\mathbb{Z}_\ell$ -module or a $\mathbb{Q}_\ell$ -vector space arising from the Tate module of an abelian variety or the étale cohomology of a variety.

APPENDIX D. SELMER GROUPS

Definition D.0.1 (Kummer map associated to an isogeny of an abelian variety). Let k be a field, and let A and B be abelian varieties (Definition A.0.5) defined over k . Suppose $\varphi : A \rightarrow B$ is an isogeny defined over k (Definition A.0.6). Denote by $G_k = \text{Gal}(\bar{k}/k)$ the absolute Galois group of k .

The short exact sequence

$$0 \rightarrow \ker \varphi \rightarrow A(\bar{k}) \xrightarrow{\varphi} B(\bar{k}) \rightarrow 0.$$

(♠ TODO: cite long exact sequence of galois cohomology) yields the following long exact sequence in Galois cohomology:

$$\begin{aligned} 0 \rightarrow \ker \varphi(k) \rightarrow A(k) \xrightarrow{\varphi} B(k) \\ \xrightarrow{\delta_\varphi} H^1(G_k, \ker \varphi) \rightarrow H^1(G_k, A(\bar{k})) \rightarrow H^1(G_k, B(\bar{k})) \rightarrow \cdots \end{aligned}$$

The *Kummer map associated to the isogeny φ* is the Galois cohomological connecting homomorphism

$$\delta_\varphi : B(k) \rightarrow H^1(G_k, \ker \varphi),$$

above. In fact, there is a short exact sequence

$$0 \rightarrow B(k)/\varphi(A(k)) \xrightarrow{\delta'_\varphi} H^1(G_k, \ker \varphi) \rightarrow H^1(G_k, A)[\varphi] \rightarrow 0$$

where the map δ'_φ is induced by δ_φ . We may also let the *Kummer map associated to φ* refer to this map δ'_φ . We may also use the abuse of notation δ_φ to denote δ'_φ .

Definition D.0.2 (Selmer group for an isogeny of abelian varieties over a global field). (♠ TODO: It should be possible to define this for more general group schemes) Let K be a global field, and let A and B be abelian varieties (Definition A.0.5) defined over K . Suppose $\varphi : A \rightarrow B$ is an isogeny defined over K .

Denote by $G_K = \text{Gal}(\overline{K}/K)$ the absolute Galois group of K . For each place v of K , fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group $G_v \subset G_K$.

The *Selmer group of φ over K* , denoted by notations such as $\text{Sel}^\varphi(A/K)$, $\text{Sel}^{(\varphi)}(A/K)$, $\text{Sel}_\varphi(A/K)$, is the subgroup of the Galois cohomology group (Definition C.0.5) $H^1(K, \ker \varphi)$ given by

$$\text{Sel}^\varphi(A/K) := \ker \left(H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A(\overline{K}))[\varphi] \right),$$

(♠ TODO: define the kummer map associated to φ) where the product runs over all places v of K , and $H^1(G_v, A)[\varphi]$ denotes the image of the local Kummer map associated to φ .

We describe the map

$$(A) \quad H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A(\overline{K}))[\varphi]$$

(♠ TODO: define a decomposition group) used to define the kernel above: Note that G_v acts on $A(\overline{K}_v)$ and $B(\overline{K}_v)$, and the base change φ_v of φ to K_v induces a Kummer short exact sequence (Definition D.0.1)

$$0 \rightarrow B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi]) \rightarrow H^1(G_v, A(\overline{K}_v))[\varphi] \rightarrow 0.$$

The natural inclusions $G_v \subset G_K$ and $E(K) \subset E(K_v)$ give restriction maps on cohomology, and we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & B'(K)/\varphi(A(K)) & \xrightarrow{\delta_\varphi} & H^1(G_K, A[\varphi]) & \longrightarrow & H^1(G_K, A(\overline{K}))[\phi] \longrightarrow 0 \\ & & \downarrow & & \downarrow & \searrow \text{dotted} & \downarrow \\ 0 & \longrightarrow & \prod_v B'(K_v)/\varphi(A(K_v)) & \xrightarrow{\delta_{\varphi_v}} & \prod_v H^1(G_v, A[\varphi]) & \longrightarrow & \prod_v H^1(G_v, A(\overline{K}_v))[\phi] \longrightarrow 0 \end{array}$$

The map (A) is the one given in the above commutative diagram by the dotted arrow.